Udemy ML, DL, NLP course (1)
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~~lec~~ sec - 34
Random Forest Machine Learning

lec 1 :- Bagging & Boosting Ensemble Techniques

lec 2 :- Random Forest regression

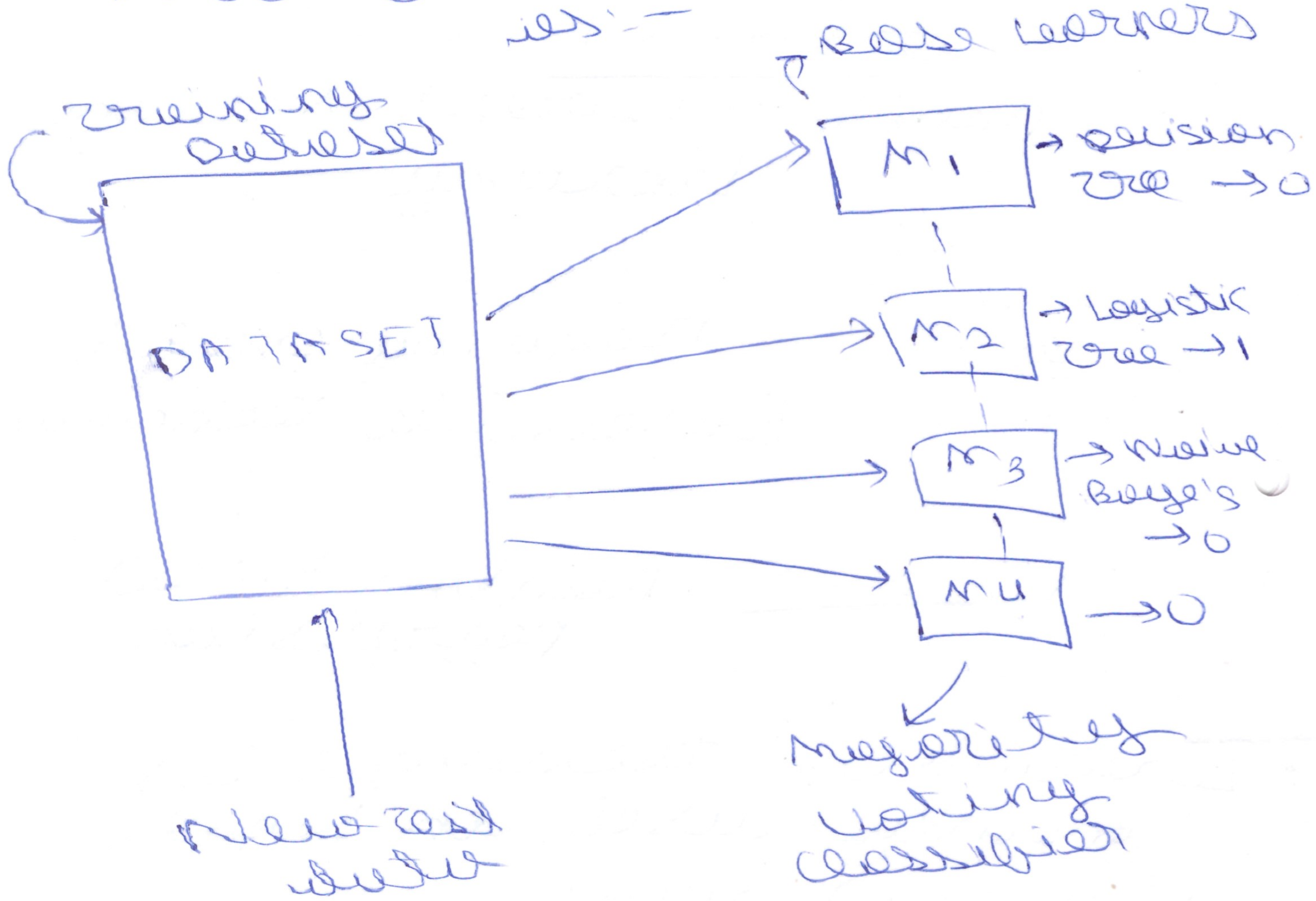
· Ensemble technique is mainly used in Kaggle & Hackathons.

Bagging :-

(1) Random Forest Algorithms

⇒ Idea behind Bagging

The main idea behind Bagging can be described as:-



Here suppose we have the training dataset which is ~~mode~~ passed through the different models.

- The final result in case of classification is the majority of all output.
- In case of Regression, its the average

→ Boosting

(1) ~~Grad~~ Adaboost

(2) ~~Xg Boost~~ Gradient Boosting

(3) Extreme Gradient Boosting (Xg Boost)

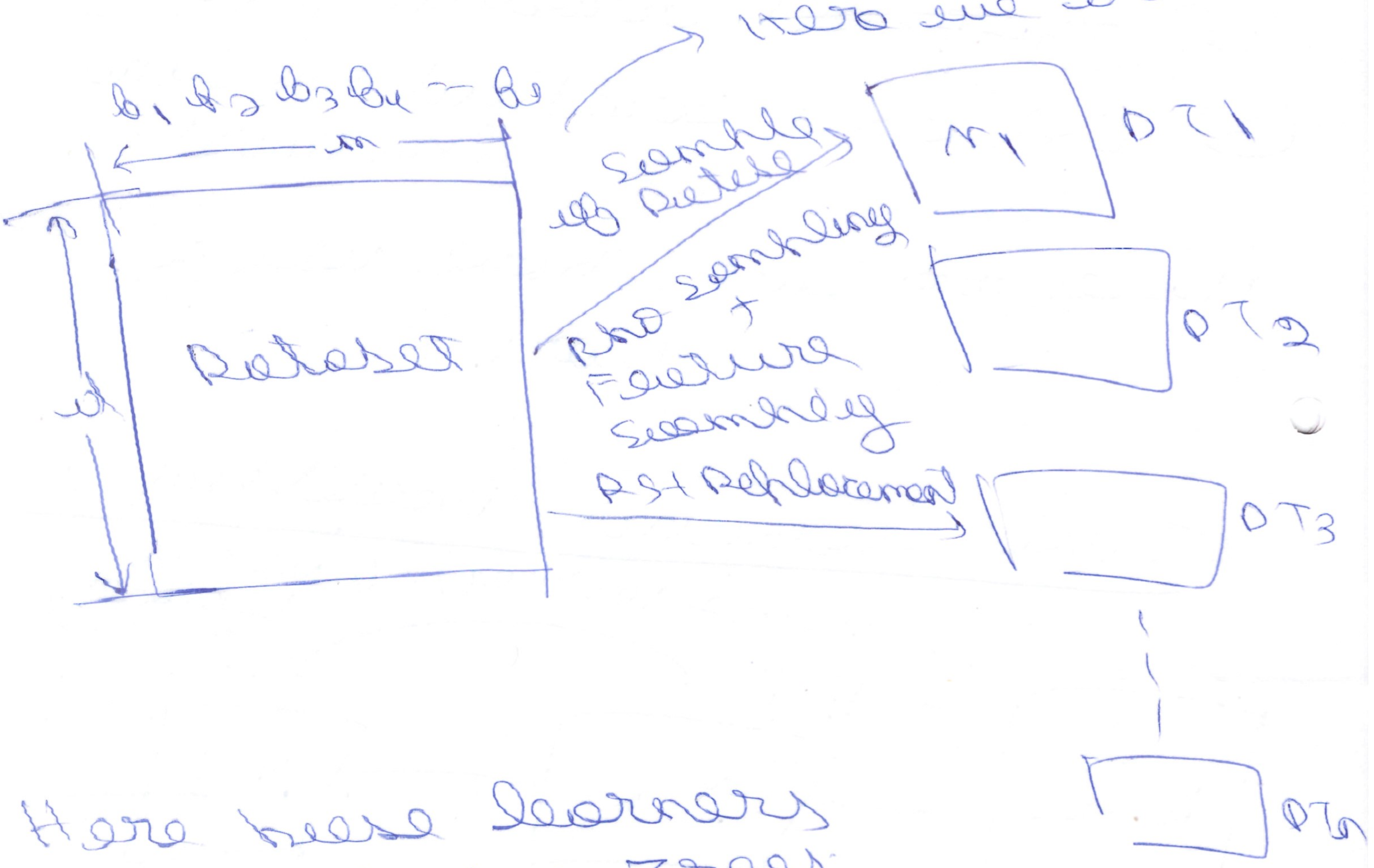
Here ~~at~~ we have weak learners

Idea behind Boosting :-

strong learner

Here the weak learner models combined to give a stronger model.

=> Random Forest Classification and regression



Here base learners are decision trees

Whenever we give new test data,

In case of DT1 → 0
" " " DT2 → 0
" " " DT3 → 0
" " " DT4 → 1

In classification we take majority vote classifier

f...

In regression we take the average of all the outcomes.

Classification - Majority voter classifier

regression → average O/P of models

⇒ Why use Random Forest instead of decision tree?

~~Classify~~

Decision Tree :-

↳ overfitting

Training Acc ↑ → Low Bias

Test accuracy ↓ → High variance

Generalized model → Low Bias

→ Low ~~Bias~~ variance

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In random Forest, we try converting the low Bias to High variance.